

Completion Guide

Everything left to do, in order, with the commands. Packetgeist — wireless field studies. Written 28 August 2026.

Where things stand

Done	Remaining
Survey executed — 2 floors, 67 points, 1,523 observations Data extracted from the .netspu database Six findings written with evidence Findings report (PDF) and data pack (xlsx) Full repository with CSVs, docs, licensing Project site built (index.html)	Export the heatmaps Crop and add the two client screenshots Build the predictive model Push to GitHub Point the domain at it Publish the two earlier projects the same way Send the report to the library

Estimated remaining effort: **one evening for the model, one evening for everything else.** The survey itself is the hard part and it is finished.

Step 1 — Export the heatmaps (20 min, phone)

- ☐ Open the project → WiFi heatmaps.
- ☐ Map menu (three dots) → **Include/exclude networks** → deselect every non-library network. This is the single biggest visual-quality win available and takes two minutes.
- ☐ Set **Min signal level to -85** and **Max to -35** in the display settings, so the colour gradient actually spans your data instead of washing everything orange.
- ☐ For each floor, cycle **Signal level**, **SNR**, **SIR** and use **Export to image** — not a screenshot. Six files.
- ☐ Also export a **neighbours-only** version: invert the network filter. That map is your external interference exhibit and almost nobody at junior level thinks to produce one.
- ☐ Name them as in assets/heatmaps/README.md and drop them in that folder.
- ☐ The SNR maps will be empty — the Android build reported no noise floor. Export them anyway and caption them as such. An honest empty map beats a silently omitted one.

Step 2 — Crop the client screenshots (5 min)

- ☐ Android Wi-Fi list showing 'Security: weak' on the public SSID — use as is.
- ☐ **Crop the Network Details screenshot to the two link-rate fields only.** The full screen shows the library's client IP, gateway, DNS and subnet mask. That is their infrastructure, not yours to publish.
- ☐ Both into assets/screenshots/.

Step 3 — Build the predictive model (one evening, desk)

This is the step that changes what the project is. Full method is in planning/README.md. The short version:

- NetSpot → Planning → load the same two floor plans, calibrate scale identically.
- Draw the walls using the materials in section 2.3 of the findings report — brick cavity over CMU for the wings, low-E curtain wall for the central volume, gypsum partitions.
- Place the eight access points at their surveyed positions with their current channels.

- **Calibrate against measurement.** If the model predicts –60 where you measured –50, adjust wall attenuation until they agree. Write down what you changed.
- Duplicate. Apply the recommended plan and render both.

What the comparison should show. Not better coverage — coverage is already 100% against threshold. The proposed model should show *fewer co-channel radios audible per location* while holding the –67 dBm floor. Saying that correctly is more impressive than claiming a coverage win that does not exist.

Step 4 — Push to GitHub

Create an empty public repository named **plainsboro-library-survey** on GitHub, then:

```
cd packetgeist-survey
git init
git add .
git commit -m "WiFi site survey: Plainsboro Public Library, Aug 2026"
git branch -M main
git remote add origin git@github.com:<you>/plainsboro-library-survey.git
git push -u origin main
```

- ☐ Repository description: 'Passive 802.11 site survey of a three-storey public library. 67 points, 1,523 observations, 8 APs. Coverage was never the problem.'
- ☐ Topics: `wifi`, `site-survey`, `802-11`, `wireless`, `netspot`, `rf`, `cwna`, `network-engineering`
- ☐ Settings → Pages → Source: **Deploy from a branch**, branch **main**, folder **/ (root)**.
- ☐ Wait two minutes, then confirm the site loads at the `github.io` URL.

Step 5 — Point the domain

You want this separate from `cherwood.tarunc.com`, so use a new subdomain on the same apex — **packetgeist.tarunc.com**. Same root domain, distinct identity, no new registration cost.

Where	What to do
DNS (your registrar / Cloudflare)	Add a CNAME record: <code>packetgeist</code> → <code><you>.github.io</code> . If you are on Cloudflare, set the proxy to DNS only (grey cloud) or GitHub's certificate provisioning will fail.
GitHub	Settings → Pages → Custom domain: <code>packetgeist.tarunc.com</code> . Tick Enforce HTTPS once the certificate is issued (can take up to an hour).
Repository	GitHub writes a CNAME file into the repo when you set the custom domain. Pull it before your next push or you will overwrite it.

Why a subdomain rather than a new domain. Zero cost, instant, and it keeps everything under one identity you already own. If Packetgeist grows into its own thing later, moving it to a standalone domain is a DNS change, not a rebuild.

Step 6 — Publish the earlier projects the same way

You already have the Cherwood Health FortiGate lab and the Kindle digest pipeline live. The goal here is that all three are browsable, consistent, and reachable from one place.

Project	Where it should live	What it demonstrates
Cherwood Health — FortiGate / AP lab	<code>cherwood.tarunc.com</code> (already live)	Routing, switching, firewall policy, segmentation. Your strongest piece for a general network engineer role.
Kindle digest pipeline	GitHub Pages on the existing repo	Automation, APIs, Linux, containers, and a security audit you performed on your own work.
Plainsboro survey	<code>packetgeist.tarunc.com</code>	RF measurement, analysis, and professional field conduct. Your strongest piece for a wireless role.

- ☐ For each repo: Settings → Pages → deploy from `main`, `root` or `/docs`.

- ☐ Add a one-paragraph README to any repo missing one, with the same structure as this survey's: what it is, what was found or built, how to reproduce it, what its limitations are.
- ☐ On tarunc.com, add a short index linking all three with one line each. That page is what you put on a resume — not three separate URLs.
- ☐ Cross-link: each project page links back to tarunc.com.

Step 7 — Send it to the library

Do this. Two staff members gave you permission; sending the finished report closes the loop and converts a practice exercise into work with a real recipient. Keep it short:

Subject: WiFi coverage findings — the survey I ran on 27 August

Hello,

I'm the graduate student who asked permission to walk the building taking WiFi signal readings on the evening of 27 August. Thank you for saying yes.

I've finished the analysis and attached the report, in case it's useful to whoever supports your network. The short version: coverage is genuinely good everywhere I measured — every location met the standard design threshold. What I did find is that several access points are sharing the same radio channels, which makes them compete with each other for airtime. The suggested fixes are configuration changes rather than new equipment.

I've noted clearly where I'm uncertain, and one finding needs someone with admin access to verify. I only listened to the public broadcast signals — nothing on your systems was accessed.

Happy to answer questions if any come up.

Tarun Cherukuri

Attach the findings PDF only. Not the data pack, not the raw export — link the site if they want more.

Step 8 — Write the process narrative

One page, on the site or LinkedIn. Hiring managers read this more closely than the report. The structure that works:

- **What you expected.** You went in looking for dead zones. That is what a beginner assumes a site survey finds.
- **What you actually found.** No dead zones at all — 100% of points met threshold. The problem was the opposite: too many access points on too few channels.
- **How you got there.** The heatmap did not show it. Opening the export as a database and grouping BSSIDs by radio prefix did.
- **What you got wrong along the way.** You trusted 'map coverage 75%, good' from six data points before working out that it measures spread, not density. Say so.
- **What you would do differently.** Photograph the APs. Record materials on site. Survey at two different times.

Do not oversell it. Describe it accurately: a self-directed practice survey with a consumer phone, conducted with the venue's permission. Interviewers in this field can tell instantly, and the accurate version is already strong for someone at your stage. You went out, did the work, respected the space, and documented it properly. Most candidates never leave their apartment.

Final check before you call it done

- ☐ No internal IP addresses anywhere in published material.
- ☐ No identifiable patrons in any image.
- ☐ No third-party architectural photography republished.
- ☐ Architectural Record credited for the floor plans and product schedule.
- ☐ F-04 still clearly labelled unverified everywhere it appears.

- ☐ The two documented exclusions still stated as scope decisions, not apologies.
- ☐ Every number on the site traceable to a CSV in data/.